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AI-powered acoustic surveillance for early detection of calf respiratory disease

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ABSTRACT

Effective management of Bovine Respiratory Disease Complex (**BRDC**) requires timely, non-invasive diagnostic tools to protect calf health and welfare. Among early clinical signs, coughing stands out as both frequent and informative. To explore its potential for early BRDC detection, we deployed an artificial intelligence (**AI**)-driven acoustic monitoring system that recorded over 2 730 h of audio during a 30-day period. Four experimental pens, each housing seven calves and stratified by infection status and antibiotic treatment, were equipped with a dedicated microphone to enable targeted acoustic surveillance. This configuration enabled pen-specific detection of cough events, which were subsequently classified using an AI lightweight Hidden-Unit BERT (HuBERT)-based model trained on 1 045 labelled clips. The classifier achieved 92% accuracy. Temporal patterns in cough frequency aligned with infection dynamics, treatment responses, and circadian patterns. Notably, AI-detected coughs consistently preceded clinical scores by 1–2 days, confirming the system's sensitivity to early respiratory disorders. These findings support the use of acoustic surveillance as a valid, scalable, and autonomous tool for continuous monitoring and early warning of respiratory diseases in calves.

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Implications

Respiratory disease remains a major challenge for calf health and welfare, creating a need for practical tools that support earlier detection than routine clinical observation. This study evaluated whether continuous acoustic monitoring could identify respiratory disease through cough detection. The results showed that automated analysis of cough sounds tracked disease progression and treatment responses, and detected changes one to two days before clinical scoring. These findings suggest that acoustic monitoring may be a useful component of precision livestock farming, supporting continuous health surveillance and earlier management of respiratory disease in calf-rearing systems.

Introduction

Bovine Respiratory Disease Complex (**BRDC**) is one of the most significant health and welfare challenges in beef cattle production worldwide (Gaudino et al., 2022a, b; Taylor et al., 2010; Timsit et al., 2016) and is also a major driver of antimicrobial use. This multifactorial syndrome arises from the interaction of environmental stressors, host-associated (holobiont) factors, and a diverse range of respiratory pathogens (Arcangioli et al., 2008; Gaudino et al., 2023b; Griffin et al., 2010; Holman et al., 2015; Klima et al., 2019; Salem et al., 2019, 2020). Young calves are particularly susceptible, as BRDC can impair immune development, induce pain and distress, and substantially increase the risk of long-term morbidity and mortality (Cummings et al., 2022). This is particularly relevant during transfer to fattening units, where calves from different origins are mixed, exposed to novel pathogens, and subjected to additional stress.

Bovine respiratory disease complex early detection is essential not only to reduce economic losses but also, more importantly, to ensure timely, targeted interventions that increase treatment success, prevent suffering, and thereby protect animal health and welfare (Nielsen et al., 2023). BRDC is clinically characterised by signs such as coughing, sneezing, nasal discharge, and tachypnoea (Carpentier et al., 2018; Gaudino et al., 2022b), with coughing being a particularly informative symptom for early detection (Carpentier et al., 2018). According to EFSA (2025), the sensitivity

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and specificity of coughing as an indicator of respiratory disorders are high (Nielsen et al., 2025). Coughing results from a forceful expulsion of air from the lungs triggered by a sudden contraction of the diaphragm and intercostal muscles in response to irritation of the lower respiratory tract (Nielsen et al., 2025). The Welfare Quality protocol (2023) for fattening cattle recommends quantifying cough frequency by continuous observations of pens or, for very large pen sizes, by monitoring subgroups of no more than 25 animals (Welfare Quality Network, 2023). However, the expansion of large-scale farming systems, combined with increasing labour constraints, makes the widespread adoption of manual coughing assessments difficult in practice. Furthermore, traditional BRDC diagnostic methods, which rely on pathogen-oriented sampling during farm visits (Werid et al., 2024), are costly, invasive, and potentially stressful, and are often performed with a delay, which limits their effectiveness in detecting early disease onset (Zhang et al., 2022).

These limitations have accelerated the development and adoption of Precision Livestock Farming technologies, which use automated data collection to support on-farm decision-making and are becoming increasingly important for early disease detection in intensive livestock production systems (Norton and Berckmans, 2018). Cough monitoring technologies have been developed to support on-farm early detection of respiratory disease in livestock. In pigs, algorithms using historical data and room-specific variations generated a respiratory distress index that correlates moderately ($r = 0.5$) with manually recorded coughing frequencies (Pessoa et al., 2021). The same study further showed that this index, measured at the end of the finishing period (21–24 weeks of age), correlated strongly ($r > 0.7$) with the prevalence of lung lesions and pneumonia assessed at slaughter. In cattle, Vandermeulen et al. (2016) reported high specificity (99.2%) and precision (87.5%) for calf cough detection using a cough-monitoring algorithm based on sound recordings, while Carpentier et al. (2018) also demonstrated the field applicability of an acoustic algorithm for detection of coughing sounds in calf house, with precision exceeding 80% and a clear correlation between cough frequency and BRDC diagnosis. Complementary approaches using collar-mounted sensors also achieved strong performance metrics, including 88.9% accuracy (Kim et al., 2024), reinforcing the viability of acoustic and behavioural monitoring for early disease surveillance.

While these results support the potential of digital farming sensors for cough as an automated tool for continuous detection of respiratory disorders in calves, further optimisation is required to improve specificity and scalability in precision livestock systems. In this context, the COVID-19 pandemic served as a powerful accelerator for acoustic cutting-edge surveillance technologies, demonstrating how AI-powered cough detection systems could be deployed at scale for early diagnosis and outbreak monitoring (Askari Nasab et al., 2023; Ayappan and Anila, 2025; Feng et al., 2024; Pentakota et al., 2023). These developments, encompassing refined cough signal processing, machine-learning and deep-learning algorithms, and remote health-monitoring infrastructure, provide a robust translational framework for veterinary medicine. Building on these recent advances in cough signal processing and AI algorithms, this study aimed to assess whether an AI-driven acoustic monitoring system can reliably detect cough events in young calves and support early diagnosis of respiratory disorders under controlled experimental conditions. To address this aim, we deployed continuous pen-level acoustic surveillance for 30 days across four experimental pens, stratified by infection status and antibiotic treatment. Calves were monitored from arrival, and respiratory pathogens were inoculated at a defined time point to standardise disease onset. Cough events captured by strategically placed microphones were systematically compared with veterinarian-performed cough assessments. This controlled design enabled precise temporal alignment between acoustic signals and the true onset of respiratory disease, strengthening the reliability and diagnostic value of cough-based monitoring systems.

Material and methods

Animal handling and groups at the experimental facility

Following birth at INRAE-Herbipole, 28 healthy and unweaned calves were transported to the national veterinary school in Toulouse (**ENVT**, Toulouse, France). At arrival (day −11), calves were 30.7 ± 13.3 days old and underwent a complete clinical examination to confirm they were clinically healthy. Animals were then balanced by BW and sex into four experimental groups: (i) infected with BRDC-associated pathogens without antibiotic treatment (***Inf_NonATB***), (ii) infected with BRDC-associated pathogens and treated with antibiotics (***Inf_ATB***), (iii) uninfected and treated with antibiotics (***NonInf_ATB***), and (iv) uninfected and untreated (***NonInf_NonATB***). This factorial design stems from the broader experimental project, which investigates how antibiotic treatment may modify the microbiome, glycome, and mucus, thereby influencing susceptibility to respiratory disease. In the context of the present study, this structure was also essential for generating distinct, biologically relevant gradients in coughing activity. By combining different infection statuses with the presence or absence of antibiotic treatment, the four groups exhibited heterogeneous coughing profiles before and after exposure to BRDC-associated pathogens, conditions necessary for rigorously validating the AI-based acoustic monitoring system.

Building on this structure, antibiotic treatment with tilmicosin (12.5 mg/kg, twice daily; Tilmovet 250 mg/mL, Huvepharma, Antwerp, Belgium) was administered from days −5 to −3 of the experiment. Before the experimental coinfection, all calves underwent comprehensive pathogen screening to ensure they were free of major respiratory agents. Specifically, from birth until arrival at the experimental farm, calves were monitored weekly for Mycoplasma bovis, bovine coronavirus (**BCoV**), and influenza D virus (**IDV**) using PCR analysis of nasal swabs (BioT respiratory qPCR kits; BioSellal, France). They were also tested for enteric pathogens (rotavirus, Giardia, Cryptosporidium), with only one calf (ID18, *NonInf_ATB*) testing positive for Cryptosporidium. Immediately prior to inoculation (Day 0), nasal swabs were analysed by real-time PCR to confirm the absence of Mannheimia haemolytica, Pasteurella multocida, Mycoplasma bovis, Histophilus somni, BCoV, IDV, bovine respiratory syncytial virus, and bovine parainfluenza virus type 3. The absence of bovine viral diarrhoea virus was verified by ELISA targeting the Erns antigen (IDEXX BVDV Ag/serum, IDEXX Laboratoires, Saint-Denis, France). Serological testing further confirmed that all calves were negative for Mycoplasma bovis-specific antibodies (Bio K 302; BioX Diagnostics) and IDV-specific antibodies (hemagglutination inhibition assay). This multi-layered screening ensured that all animals entered the study free of major respiratory pathogens, providing a clean baseline for evaluating the onset of respiratory disease under controlled infection conditions.

On day 0, calves assigned to the infected groups were co-inoculated with three BRDC-associated respiratory pathogens: IDV (strain D/bovine/France/5920/2014), BCoV (strain ENVT-66), and Mycoplasma bovis (strain RM16) (Lion et al., 2021). Inoculation was performed via intranasal nebulisation (8–12 min) using a foal-adapted mask and compressor system, delivering 10^7 $TCID_{50}$ of IDV and BCoV and 10^{10} CFU of Mycoplasma bovis per calf.

At days +1, +3, +7, +9, +14, and +19 following infection, animals were screened for *Mycoplasma bovis*, IDV, and BCoV using nasal swabs and bronchoalveolar lavage (**BAL**) fluid. Viral RNA and bac-

terial DNA were extracted from 170-µl aliquots of nasal swab medium, raw BAL fluid, and faecal swab samples using the NucleoMag Pathogen kit (Macherey-Nagel, Germany) on the KingFisher Flex purification system (Thermo Fisher Scientific, MA, USA) according to the manufacturer's instructions. The PB1 (polymerase basic 1) gene of IDV, the N (nucleocapsid protein) gene of BCoV, and the oppD (oligopeptide permease D) gene of Mycoplasma bovis were amplified using specific primers and quantified using the iTaq Universal SYBR Green One-Step kit (Bio-Rad Laboratories, USA) on a LightCycler 96 Real-Time PCR System (Roche Diagnostics, Germany). The bovine GAPDH gene was used as an internal control to assess sample quality and the efficiency of nucleic acid extraction. Copy numbers in samples were determined using standard plasmids containing the PB1 product of D/bovine/France/5920/2014 (IDV), the N product of strain ENVT-66 (BCoV), or the oppD product of strain RM16 (Mycoplasma bovis).

In parallel, at the same time points, nasal swabs and BAL samples were analysed using the commercial BIO-T Bovine Respiratory Syncytial Virus, Parainfluenza Type 3, M. bovis, *H. somni*, M. *haemolytica* and P. *multocida* PCR kits (Biosellal, France), which detect the major pathogens involved in the bovine respiratory disease complex: *Mannheimia haemolytica, Pasteurella multocida, Histophilus somni*, Mycoplasma bovis, bovine respiratory syncytial virus, and parainfluenza virus type 3.

The study took place in a biosafety level 1 barn (35 × 11.3 × 5 m) with natural ventilation and four pens (4.9 × 14 m; 10.5 m^2/-calf), arranged in two rows separated by a 1.5 m corridor (Fig. 1AB). Each pen housed seven calves and featured open sides, solid rear walls, and a galvanised mono-pitch roof. Infected and uninfected groups were separated by a transparent polyethylene barrier, allowing visual but not tactile contact. Pens operated independently under a "*marche en avant*" protocol, with no shared equipment or personnel crossover. Bedding (pine shavings) was spot-cleaned daily and replaced every two days.

Milk replacer was offered twice daily (0700 and 1700 h), with a 30-minute feeding window. The quantity of milk refused was recorded without force-feeding. The milk replacer (Univor, *Spécial Élevage*, France) was adjusted weekly to achieve a target growth rate of 0.8 kg/day. Initial volumes ranged from 3–4.5 L per meal and increased to 4.7–6.7 L based on BW. Calves had *ad libitum* access to fresh water via automatic drinkers. Neither concentrate nor hay was provided during the study to maintain strict control over nutrient intake and ensure that all calves received a standardised, homogeneous diet throughout the experimental period. Solid feeds introduce substantial variability in energy, fibre, and micronutrient intake, which can differ markedly between individuals depending on appetite, hierarchy, and exploratory behaviour. Such variability could have introduced uncontrolled differences in physiological responses, clinical expression, and coughing activity, thereby complicating the interpretation of the effects of infection and antibiotic treatment on respiratory outcomes and on the performance of the AI-based acoustic monitoring system.

Clinical cough assessment by veterinarians

Calves were examined from arrival (Day −11) to 19 days postinfection, following a standardised clinical scoring protocol adapted from Lion et al. (2021) and consistent with the clinical assessment

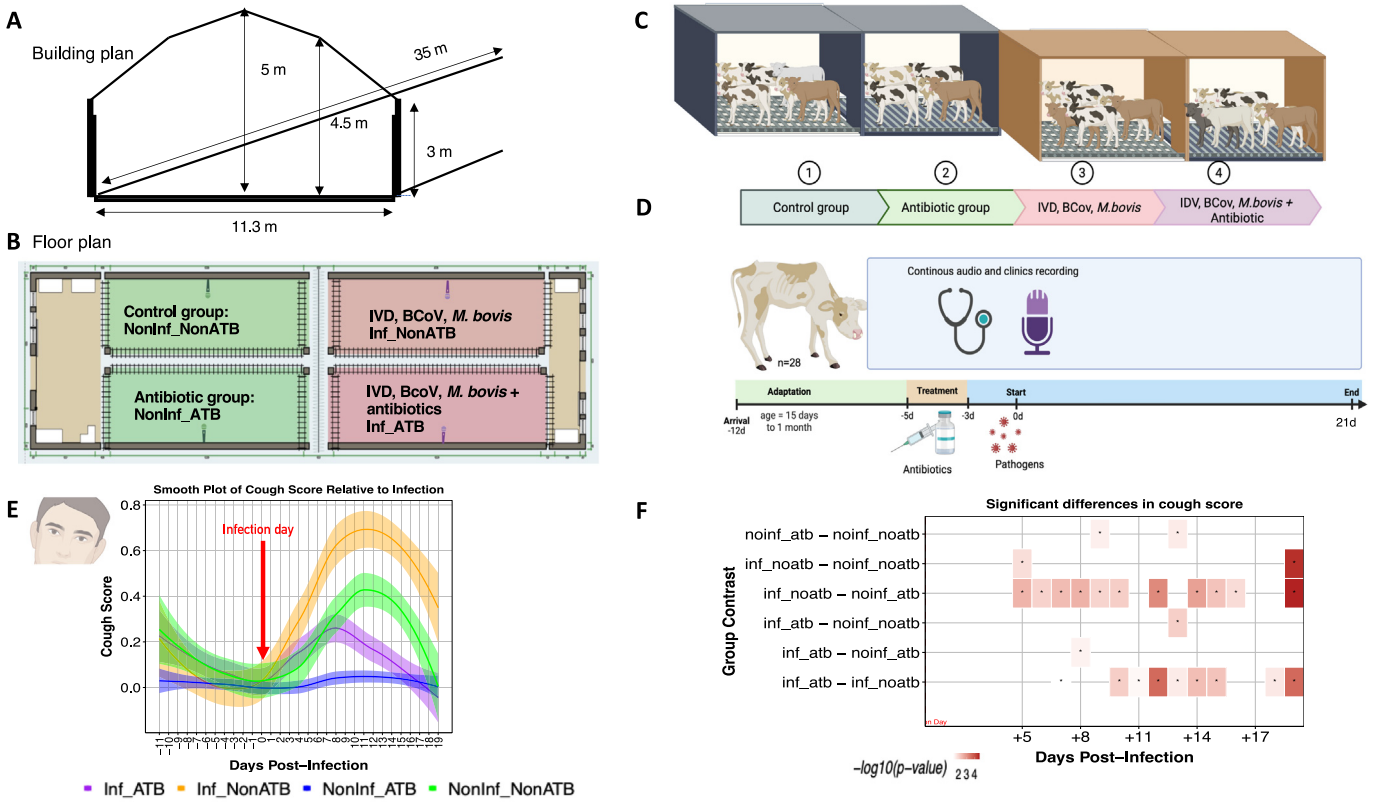


Fig. 1. Experimental Setup and Clinical Cough Monitoring in Calves. (A–B) show pen layout and group allocation: 28 calves were divided into four groups, as defined in Table 1. Each group was housed in a microphone-equipped pen for continuous cough monitoring; (C) summarises pathogen exposure (influenza D virus (IDV), bovine coronavirus (BCoV), and Mycoplasma bovis) and antibiotic protocol; (D) Project timeline: calves arrived 12 days before infection (Day 0), with monitoring continuing until Day 19 via twice-daily clinical scoring and uninterrupted audio recording; (E) Smoothed mean cough scores by day postinfection (Day 0), shown for each experimental group; and (F) Daily pairwise comparisons highlight significant group differences. Tile colour intensity reflects $-\log_{10}(P\text{-value})$, with darker reds indicating stronger significance; asterisks mark $P < 0.05$. Pale or grey tiles denote non-significant or missing data.

principles described in Meyer's group (Arcangioli et al., 2008; Gaudino et al., 2023a; Meyer et al., 2023; Salem et al., 2020). During each examination, the veterinarian evaluated general condition, appetite, rectal temperature, nasal discharge, respiratory effort, cough, and lung sounds. Thoracic ultrasonography was also performed on all calves throughout the experiment. All lung lobes were systematically examined using a standardised scanning protocol, and all animals were confirmed to be free of detectable pulmonary lesions prior to the experimental coinfection. Although the complete clinical score was recorded for the broader experimental project, only the cough-related variable was retained to validate the AI-based acoustic monitoring system.

Cough was defined as a spontaneous and audible expulsion of air from the mouth, a disease-related event occurring while the calf was quietly standing or lying at rest, without external triggers such as activity, agitation, dust, feed particles, or handling. Each calf was first observed at rest for a defined period to identify spontaneous coughs; auscultation was performed afterwards to assess other respiratory signs. Only coughs recorded prior to auscultation were considered for classification. Abnormal tracheal or lung sounds detected during auscultation were recorded for clinical monitoring but were not used to classify an animal as coughing, as these sounds are not detectable by the AI system and do not correspond to cough events. Spontaneous coughs were classified solely by frequency, following Meyer's BRDC phenotyping approach, with scores of 0 (no spontaneous cough observed), 1 (one or a few isolated spontaneous coughs), or 2 (multiple spontaneous coughs or short bouts). For comparison with AI detections, this ordinal score was converted into a binary variable, where score 0 corresponded to cough absent, and any score of 1 or 2 corresponded to cough present, meaning that a single disease-related spontaneous cough was sufficient to classify the calf as coughing.

To ensure consistency and biosecurity, veterinarians were assigned exclusively to either infected or uninfected pens, with no personnel interchange throughout the study. Each calf was evaluated twice daily by two veterinarians working independently within their assigned area, providing multiple annotations per day to minimise observer bias and ensure that all labelled cough events corresponded to true disease-related spontaneous coughs.

Environmental data

Automated sensors (VelociCalc® 9565-A, TSI, USA) recorded temperature, humidity, heat index, and dew point every 10 min in each pen.

Dew point thresholds (< 10 °C and > 21 °C) were selected based on established biometeorological classifications of humidity stress in cattle (Mader et al., 2006). Heat index thresholds (> 27 °C "Caution" and > 32 °C "Extreme Caution") followed NOAA/National Weather Service categories, which closely correspond to ruminant heat stress thresholds reported for dairy and beef cattle (Cartwright et al., 2023; Polsky and von Keyserlingk, 2017).

Cough dataset collection using audio

Sound was recorded with dynamic supercardioid microphones (Behringer XM1800S, Willich, Germany) fitted with foam windscreens to protect against wind and dust and to allow quick hygienic replacement. Each of the four experimental pens was equipped with a dedicated microphone, and all four groups were recorded simultaneously. Microphones were hung from the ceiling, about one metre above the calves' height. The placement of microphones in each pen aimed to maximise their separation while maintaining sufficient distance from walls to reduce unwanted sound reflections. A Raspberry Pi 4 Model B (Cambridge, UK) enabled real-time remote monitoring of input levels. Microphone signals passed through a Focusrite preamp/interface (Scarlett 2i2 USB Audio Interface, Wycombe, UK) and were captured on a ZOOM H4n Pro Handy Recorder (Zoom Corporation, Tokyo, Japan) following protocols established in previous related studies (Alsina-Pagès et al., 2021; Miranda et al., 2024; Vidana-Vila et al., 2023). Audio recordings were performed continuously throughout the experimental period, rather than during selected time windows. Monitoring extended from calf arrival before infection through the postinfection follow-up period, covering approximately 31 experimental days, from Day −11 to Day +19 relative to infection. This continuous recording strategy generated 2 730.7 h of audio across the four sensors.

Preprocessing of the audio-based cough pipeline

From audio recordings, a dataset for cough/non-cough classification was built. Eighty manually labelled clips were first used to train an initial classifier. Manual labelling refers here to acoustic labelling of recorded audio clips, not to direct in-pen observation of the calves at the time of each event. An initial set of 80 audio clips was labelled by listening to the recordings and by visually inspecting waveform and spectrogram representations using Audacity. This labelling approach followed protocols previously developed and applied by our group in related acoustic-monitoring studies (Alsina-Pagès et al., 2021; Miranda et al., 2024; Vidana-Vila et al., 2023). Labelling was carried out jointly by acoustic signal-processing engineers and veterinarians: engineers performed the segmentation and preliminary tagging of candidate acoustic events, while veterinarians supervised the procedure and validated whether each event corresponded to a cough or to a non-cough barn sound. Whenever a candidate event was acoustically ambiguous or could be mistaken for another barn sound, including vocalisations, metallic noises from gates, fences, or feeding equipment, ventilation, footsteps, or management-related sounds, the event was reviewed jointly by the engineering and veterinary teams, and the final label was assigned by consensus. Thus, the labelled dataset used for model training and evaluation reflected the acoustic information available to the AI system while incorporating veterinary expertise to define and validate cough events. Model predictions on new segments were then used to accelerate annotation: high-confidence positives and negatives were flagged for human verification. This semi-supervised loop-training, inference, assisted labelling, and dataset expansion- was repeated multiple times, increasing both the quantity and diversity of labelled data and steadily improving model performance and annotation efficiency.

All audio files were preprocessed using the Torchaudio library (Hwang et al., 2023). Recordings at 48 kHz were resampled to 16 kHz to match the pretrained lightweight Hidden-Unit BERT (**HuBERT**) front end (Hsu et al., 2021). To improve robustness under field conditions, notch filtering was optionally applied to attenuate power-line interference at 50 Hz and low-order harmonics (e.g., 100 Hz and 150 Hz). Band-pass filters (Q = 3) were implemented using forward–backward (zero-phase) IIR filtering to prevent phase distortion. When augmentation was enabled, both the original and the filtered versions of each clip were retained for training, and the train/test split was performed at the source-file (or pen/day) level to prevent leakage between sets.

Audio-based cough feature extraction and lightweight HuBERT-based architecture

The pretrained Facebook HuBERT-base-ls960 model was employed, a self-supervised speech representation architecture that generates 768-dimensional embeddings directly from raw audio waveforms. HuBERT is particularly well-suited for cough

event detection because it learns acoustic structure, enabling it to capture subtle temporal and spectral patterns characteristic of respiratory sounds. Unlike traditional handcrafted features, HuBERT embeddings encode rich, hierarchical representations of both short-term phonetic cues and longer-range acoustic dynamics, which enhances the model's ability to discriminate coughs from background noise, vocalisations, and other barn-level acoustic events. Embeddings were obtained by averaging the final encoder layer over time and fed to a lightweight classifier. Freezing HuBERT reduced computational cost and mitigated overfitting, given the limited volume of labelled data.

Model architecture of the audio-based cough classification, training strategy and inference of cough events

The classifier operated directly on the 768-dimensional HuBERT embeddings. To minimise overfitting, we used a single fully connected linear layer mapping to two classes (cough vs non-cough). A softmax function converted outputs to class probabilities; only the classifier's parameters were updated during training. This minimalist design enabled rapid iteration within the semi-supervised workflow, in which model predictions assisted in progressive data labelling and dataset growth. The training was performed using cross-entropy loss and the Adam optimiser.

During inference, WAV files are segmented into non-overlapping 1-s windows. Segments were converted to tensors with Torchaudio, afterwards resampled to 16 kHz, then normalised with the Wav2Vec2FeatureExtractor, and passed through the frozen HuBERT encoder (eval mode) to generate 768-dimensional embeddings. The trained linear classifier outputs class probabilities: segments were labelled "cough" when the predicted class was "*cough*", and the associated probability exceeded a predefined threshold (default: 0.50). Predictions were saved as timestamped text files for each input audio. The pipeline supported batch processing of user-specified folders and runs on CPU, Apple Silicon (MPS), and CUDA-enabled GPUs.

Performance assessment of the HuBERT cough detection model from audio-based data

The final model was trained and evaluated on a fully labelled set of 1 045 clips (737 non-cough; 308 cough). All clips were manually inspected to ensure label quality. The reference method used for performance evaluation was the manually labelled acoustic dataset described above. Direct in-pen observation was not used as the event-level reference method because continuous, synchronous observation of all recorded events across four pens and throughout the full experimental period was not feasible. Therefore, true positives, false positives, true negatives, and false negatives were computed by comparing model predictions against the acoustic ground truth obtained through manual labelling of recorded clips, with veterinary validation of cough and non-cough events. Standard metrics, precision, recall, F1-score, accuracy, and macro/weighted averages, computed per class and overall, were reported to ensure a balanced assessment of performance.

Precision, defined as True Positive/(True Positive + False Positive), is the proportion of predicted positives that are true and was computed separately for the cough and non-cough classes. Higher precision indicates fewer false positives. Recall (sensitivity), defined as True Positive/(True Positive + False Negative), measures the proportion of actual cough events correctly identified. Higher recall corresponds to detecting a larger share of true coughs. These metrics were computed for both classes to ensure balanced evaluation across the dataset. The F1-score, which harmonises precision and recall, was used to summarise overall classification performance.

Acoustic alarm threshold estimation and visualisation from audio-based data

To detect anomalous increases in cough activity, an alarm threshold was set using preinfection acoustic data. For each microphone, the baseline cough activity was calculated using event counts recorded during the five days preceding the infection date (Day 0). Specifically, the baseline was computed as the mean number of daily cough events plus two SD ($\mu + 2\sigma$), where μ is the mean baseline cough count (Days –5 to 0), and σ is the corresponding SD. Non-cough barn sounds associated with management activities at the site, including metallic noises from gates, fences, and feeding equipment, ventilation, footsteps, milk distribution, clinical handling, and other routine activities, were not excluded from the recordings. Instead, these sounds were explicitly considered during manual labelling and were assigned to the non-cough class when appropriate. This ensured that the classifier was trained against the acoustic variability of the barn environment and learned to discriminate cough events from other background sounds rather than from an artificially clean acoustic context. In addition, a conservative posterior-probability threshold of 0.85 was applied for cough classification, providing an additional safeguard against false-positive detections from ambiguous non-cough sounds. Daily cough event counts were then compared against this threshold. Days exceeding the threshold were flagged as alarm-positive, indicating potential early respiratory distress. These alarm flags were visualised using time-series plots, with the *x*-axis representing experimental days relative to infection, and the *y*-axis showing event counts. A red dashed line denoted the alarm threshold, while alarm-positive days were highlighted with red markers and annotated as "Alarm". All plots used a fixed *y*-axis scale to facilitate cross-sensor comparison. This approach enabled biologically relevant and interpretable detection of acoustic anomalies, supporting early intervention strategies in precision livestock monitoring.

Statistical analysis: temporal dynamics of cough frequency across treatment groups

Temporal changes in cough frequency were assessed using two complementary datasets: clinical scores from veterinarian evaluations and event-level predictions from the automated audio-based pipeline.

Veterinarian-assessed cough scores

Statistical analyses were performed in R to evaluate group-wise differences over time. A linear mixed-effects model was applied to the full dataset using the *lmer()* function, with cough score as the dependent variable, treatment group as a fixed effect, and animal ID as a random intercept to account for repeated measures. To explore daily contrasts, the dataset was grouped by day, and separate linear models were fitted for each subset. Posthoc pairwise comparisons were conducted using the emmeans package (v1.10.0) and Tukey's adjustment for multiple comparisons.

Audio-inferred cough events

Each audio-detected event was timestamped and associated with a continuous probability score. For statistical modelling, each row represented a discrete event, and daily counts per sensor were aggregated using the *aggregate()* function in R. The resulting dataset included total event counts (Event_Count) per date and group. To assess temporal and group-wise differences in cough detection, a generalised linear model with a Poisson distribution and log link function was fitted. Predictor variables included treatment group and recording day, with the reference group set to NonInf_NonATB. Statistical significance was evaluated using *z*-statistics and associated *P*-values, with a threshold of $P < 0.05$.

Comparative analysis of cough detection: audio-based vs veterinarian observations

To evaluate the performance of the AI-powered acoustic monitoring system, we compared the temporal patterns of coughing obtained from two independent sources: (i) twice-daily veterinarian assessments of spontaneous, disease-related coughs at the individual-calf level, and (ii) continuous audio-based cough detections recorded at the pen level. Because these datasets differ in scale and temporal resolution, both were aggregated to the pen × date level to allow a meaningful comparison of group-level coughing dynamics. For each pen and each day, veterinarian observations were converted into a pen-level clinical cough score, calculated as the proportion of calves classified as "cough present". This proportion was then rescaled to match the numerical range of the audio-derived cough event frequency, enabling both measures to be jointly visualised and statistically analysed.

To explore the relationship between scaled veterinarian cough scores and cough audio-based event frequency, a generalised additive model was fitted using restricted maximum likelihood (REML). The generalised additive model framework was selected because it allows flexible, non-linear relationships between the two measures without imposing a predefined functional form. The model took the form:

$$\text{Event_Frequency} \sim s(\text{Scaled_Veterinarian_Score})$$

where Event_Frequency represents the daily audio-based cough event frequency for each pen, and Scaled_Veterinarian_Score represents the rescaled pen-level veterinarian cough score. A Gaussian family with an identity link was used because the outcome variable is continuous. Model performance was evaluated using adjusted R^2, deviance explained, and the significance of the smooth term. This approach allowed us to determine whether days with higher veterinarian-observed coughing in a pen corresponded to days with higher audio-detected cough activity, providing a robust, statistically grounded validation of the AI system.

Results

Cough frequency trajectories highlight early onset, peak severity, and antibiotic-driven reduction in bovine respiratory disease complex

The progression of cough scores following infection and antibiotic administration was evaluated by veterinarians from Day −11 to Day +19 postinfection across the four experimental groups. Smoothed trajectories of mean cough scores revealed distinct temporal patterns linked to infection status and treatment (Fig. 1EF, Group, $F_{1,1\ 672}$ = 62.16, $P < 0.001$; Day, $F_{1,1\ 672}$ = 62.54, $P < 0.001$, and their interaction Group × Day, $F_{3\ 115}$ = 24.58, $P < 0.001$). Respiratory coughs emerged around Day +5 in experimentally infected calves without antibiotic treatment (*Inf_NonATB*), marking the earliest onset among all groups. Cough scores continued to rise, peaking on Day +11, reflecting the infection's most pronounced clinical expression. In contrast, infected calves treated with antibiotics (*Inf_ATB*) demonstrated a consistent therapeutic response. From Day 9 onward, their cough scores were significantly lower than those of untreated infected calves (e.g., Day 9: estimate = −0.57, P = 0.0366), confirming the efficacy of antimicrobial intervention (Fig. 1F).

Unexpectedly, the *NonInf_NonATB* group—intended as untreated, uninfected controls—also displayed notable episodes of coughing, particularly in the mid-phase of the study, albeit less pronounced than in the *Inf_NonATB* group. This finding suggests unintended pathogen exposure, potentially through cross-contamination or natural transmission from experimentally infected animals. Despite the implementation of strict biosecurity measures and physical separation between groups, both Mycoplasma bovis and BCoV were detected in initially uninfected calves approximately three days after their detection in the infected groups. This consistent temporal lag supports the hypothesis of within-facility transmission dynamics, likely mediated by indirect contact pathways.

Non-infected calves receiving antibiotics (*NonInf_ATB*) nevertheless maintained consistently low respiratory scores throughout the observation period.

Beyond infection-related dynamics, most groups exhibited an early surge in cough events during the adaptation period. This transient peak likely reflects stress associated with the initial mixing of unfamiliar animals, balanced by weight, sex, and breed. Such regrouping is known to elevate activity levels, including agonistic interactions, vocalisation, and respiratory effort, all of which may contribute to increased coughing (Lyu et al., 2023). Additionally, initial exposure to fresh pine shavings may have caused airway irritation from dust, which has been previously linked to coughing in housed calves (Donlon et al., 2023). Importantly, this transient increase in cough activity was not associated with variations in ambient temperature or humidity, which remained stable throughout the study period (Supplementary Figure S1), nor with dietary changes, as animals were maintained on a consistent milk replacer without exposure to hay or concentrate. Additionally, calves were initially free of major respiratory pathogens, further supporting that this early peak in coughing was not driven by infectious processes.

High-performance acoustic cough classification using lightweight HuBERT-based architecture

Over the study period, the four sensors recorded 2 730.7 h of audio across four calf pens and ∼30 days (Fig. 2A). Of the 482 files generated, with an average duration of 5.7 h per file, 89.0% were successfully processed. Only 53 were discarded due to corruption or absence of detectable events.

This extensive deployment yielded a large, acoustically diverse dataset that served as the basis for supervised model training and performance evaluation. After preprocessing and quality control, we developed a binary classifier using 1 045 manually annotated audio clips (308 cough; 737 non-cough). Training over 75 epochs (batch size = 16; learning rate = 0.001) resulted in rapid convergence, with loss decreasing from approximately 32.5 to 15 and the F1 score stabilising near 90% (Fig. 2B). The final model demonstrated strong performance: overall accuracy reached 0.92, with macro and weighted F1 scores of 0.91 and 0.92, respectively. For non-cough events, precision, recall, and F1 score were 0.94, 0.95, and 0.95, while for coughs, all three metrics reached 0.87, indicating high sensitivity despite substantial acoustic variability and an imbalanced class distribution (Fig. 2C). A posterior-probability threshold of 0.85 was selected for cough classification (Fig. 2D), providing a biologically meaningful and conservative cutoff for event detection.

To mitigate class imbalance, we incorporated a class-weighted loss function during training. The strongly weighted performance metrics further suggest that the inherent acoustic distinctiveness of cough events helped counterbalance the unequal class representation.

Building on this validated model, we applied the classifier to the full dataset, comprising 35 786 processed audio segments. Of these, 6 868 (19.2%) were classified as coughs, corresponding to an average of 221.5 coughs per day and 2.5 coughs per hour, while 28 918 (80.8%) were classified as non-coughs. These results confirm the classifier's ability to robustly distinguish coughs from background acoustic signals, including frequent metallic noises

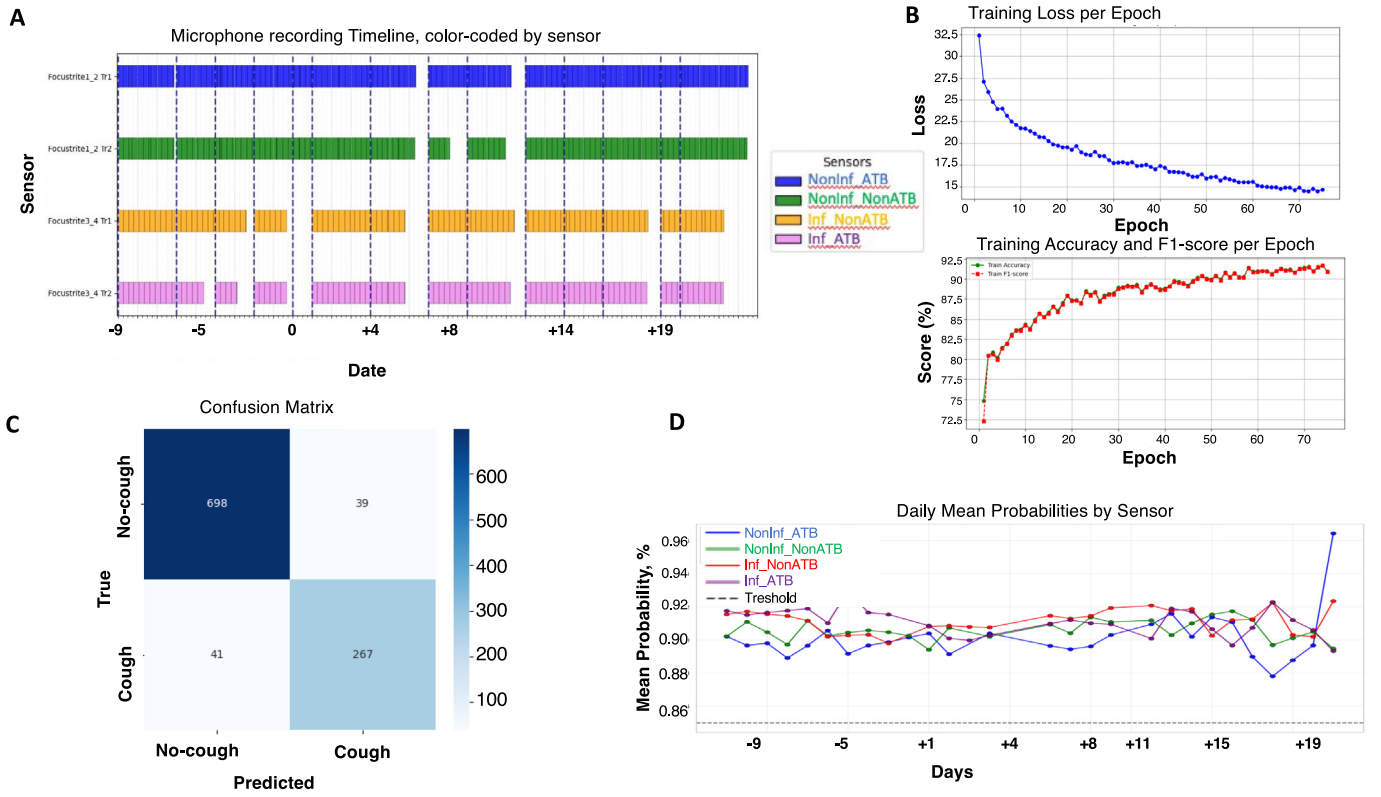


Fig. 2. Performance and deployment metrics of the acoustic monitoring system to detect cough in calves. (A) Total hours of audio recorded by each sensor across the 31-day experimental period. Bars represent cumulative recording time per sensor and group, aligned to the experimental timeline. Groups are the same as defined in Table 1; (B) Training dynamics of the binary classifier. Top: training loss per epoch; Bottom: training accuracy and F1-score per epoch. Both plots span 75 epochs (*x*-axis: 0–70 shown for clarity), illustrating rapid convergence and stable performance; (C) Confusion matrix displaying the performance of the sound classification model for detecting cough events from microphone recordings. Rows represent true labels, and columns indicate predicted classes. Values indicate the number of true and predicted labels for cough and non-cough events; and (D) Mean posterior probability of cough classification across all processed files, stratified by sensor. Lines represent the average model confidence for positive predictions, reflecting consistency across deployment sites.

generated by gates, fences, and feeding equipment within the calf rearing environment.

Acoustic cough surveillance between pens: high spatial accuracy and biological sensitivity

To evaluate group-level differences and temporal dynamics in acoustic signal reliability, cough activity was analysed across pens and time points. The classifier's overall performance was first assessed across the groups. Event frequency varied considerably across groups, whereas detection performance was consistent across sensors, with each reporting a mean confidence above 90% (Table 1).

Sensor *Inf_NonATB* recorded the highest proportion of events (32.8%), while *NonInf_ATB* reported the fewest events (17.8%; Table 1). To further assess the relationship between pen-level infection status and recorded cough frequency, a Poisson regression model was fitted using event count as the outcome variable. Using the *NonInf_NonATB* category as the reference, all groups showed statistically significant differences in cough event frequency. The *Inf_NonATB* exhibited the highest activity, with 67% more detected events ($\beta = 0.51$; $\exp^{(\beta)} = 1.67$). Conversely, *NonInf_ATB* sensors registered 44% fewer events compared to the reference ($\beta = -0.59$; $\exp^{(\beta)} = 0.56$). A significant temporal trend was also observed, with each additional day from day 11 associated with a 3.5% reduction in event detection ($\beta = -0.036$; $\exp^{(\beta)} = 0.965$), reflecting recovery over time. All effects were highly significant ($P < 0.001$).

Temporal analysis revealed notable day-to-day variation (SD = 141.7 events/day), with the highest cough activity observed on 9 days postinfection (522 events) and the lowest on the last day of the experiment (30 events; Fig. 3A). Particularly, for experimentally infected groups, the highest counts were recorded by sensor *Inf_NonATB* on Day 9 postinfection (247 events) and sensor *Inf_ATB*

Table 1
Sensor-level analysis of the HuBERT cough-detection model in calves.

Sensor ID	Group	Total events	Daily average (±SD)	Mean confidence	Confidence range
1_2_Tr1	*NonInf_ATB*	1 224	45.3 ± 44.9	90.4%	85.0–99.4%
1_2_Tr2	*NonInf_NonATB*	1 788	66.2 ± 46.0	90.7%	85.0–99.6%
3_4_Tr1	*Inf_NonATB*	2 251	83.4 ± 55.3	91.3%	85.0–99.6%
3_4_Tr2	*Inf_ATB*	1 605	59.4 ± 56.4	91.2%	85.0–99.6%

Abbreviations: Four experimental groups were included: *NonInf_ATB* = uninfected animals treated with antibiotics; *NonInf_NonATB* = uninfected animals not treated with antibiotics; *Inf_NonATB* = infected animals not treated with antibiotics; and *Inf_ATB* = infected animals treated with antibiotics.

on Day 8 postinfection (189 events; Fig. 3AB). Non-infected groups also showed periodic spikes, notably on days 13–14 postinfection (226 and 208 events, respectively). During this process, a technical anomaly was identified on day 12 postinfection (Fig. 3B). On this date, one calf in the *NonInf_NonATB* group remained inadvertently attached to the mobile feeding fence used for milk distribution, producing continuous mechanical and metallic sounds that dominated the acoustic recordings. These non-biological signals likely interfered with the classifier's ability to accurately detect cough events (Fig. 3B).

Similar to the veterinarian-based assessments, acoustic monitoring detected an increase in cough activity during the initial days of the experiment. As this rise preceded pathogen inoculation, it is unlikely to reflect respiratory disease and instead aligns with management-related factors, such as regrouping of unfamiliar animals and exposure to fresh bedding. Despite these early, non-infectious signals, the system consistently preserved clear pen-level resolution, capturing distinct spatial and temporal patterns in cough frequency associated with infection status, treatment, and time.

Interestingly, event detection peaked during early morning and late evening hours for most sensors (Fig. 3C). These event trends, visualised with smoothed lines and shaded confidence intervals, likely echoed patterns of animal behaviour. Overall, minimal event activity was observed between 0000 and 0300 h, aligning with expected resting phases.

Acoustic monitoring as an early-warning tool: evidence for predictive cough detection prior to clinical onset

To assess the alignment between AI-based acoustic monitoring and detection of veterinarian cough scores, we applied generalised additive models across all treatment groups. The temporal trajectory of cough scores assigned by veterinarians did not significantly correlate with the acoustic signal patterns in any group, including *NonInf_NonATB* (edf = 1, F = 0.011, P = 0.919), *Inf_NonATB* (edf = 1.72, F = 0.915, P = 0.392), *NonInf_ATB* (edf = 1.47, F = 0.411, P = 0.675), and *Inf_ATB* (edf = 1.00, F = 0.002, P = 0.963). Notably, in the *Inf_NonATB* group, microphone-detected cough activity began rising 1–2 days prior to cough score escalation, indicating that acoustic signals may precede observable clinical signs.

Applying a threshold-based approach to preinfection cough activity revealed clear deviations from baseline patterns. Using a mean + 2 SD criterion (average threshold ≈112 coughs/day), periods of elevated cough frequency were identified as alarm-positive events. Following infection, these alarms were first detected on Days 7 and 8 in the *Inf_ATB* group and more persistently in the *Inf_NonATB* group (Days 7, 8, 11, and 12), consistent with greater disease progression in untreated animals.

Notably, alarm-positive days were also detected in the non-infected groups on Days 12 and 13. These findings support the potential of acoustic surveillance to detect early behavioural indi-

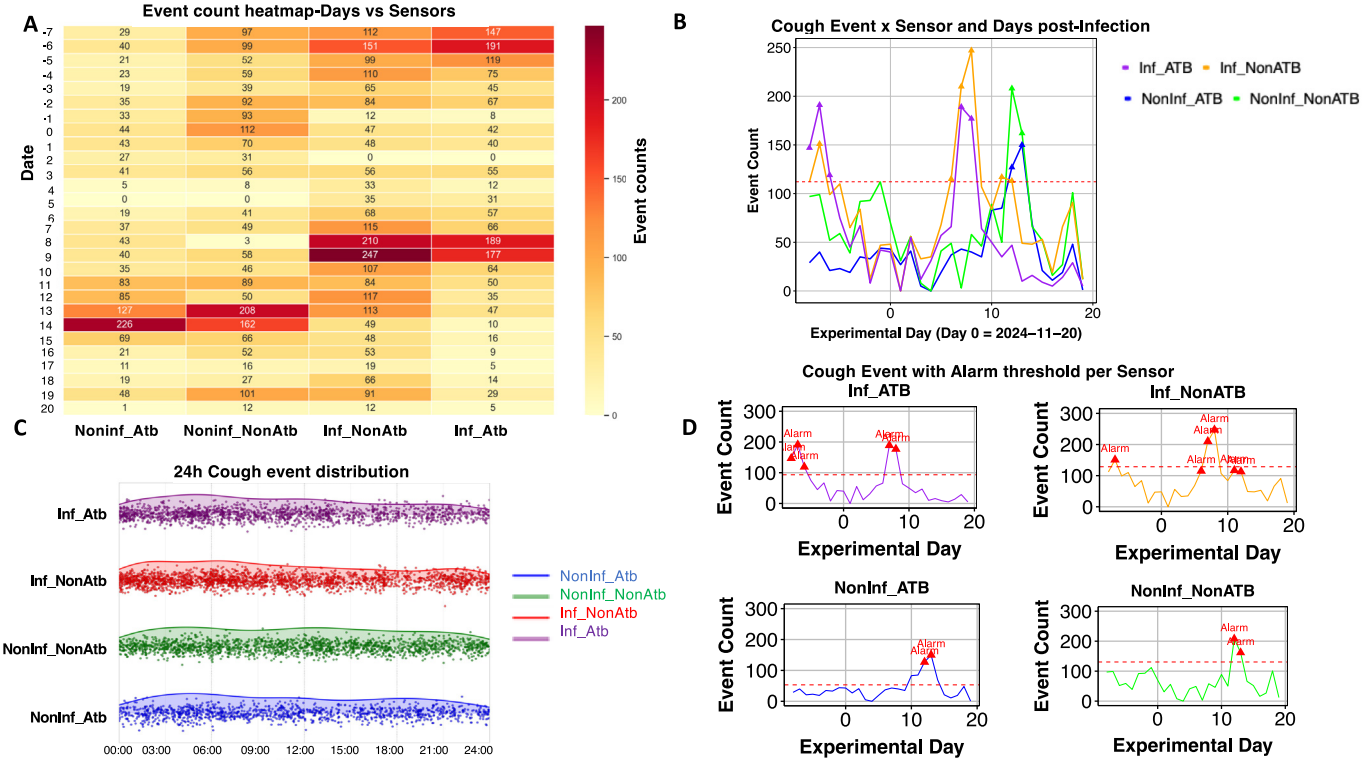


Fig. 3. Calves Cough Score and Audio Event Frequency Aligned by Days Postinfection. (A) Event Count Heatmap depicting daily detection frequencies across four sensor channels (experimental groups) using a classification threshold of ≥0.85. Darker shades indicate higher event counts. This visual representation allows for temporal and spatial comparison of sound event detection; (B) Temporal patterns of audio event frequency, aligned by days postinfection across four experimental groups; (C) Circadian pattern analysis of cough detection events with kernel density smoothing. Individual events are plotted as scatter points along the 24-h timeline (*x*-axis) for each sensor (*y*-axis), with random vertical jitter to prevent overlap. Coloured trend lines show kernel density estimates of activity patterns, with filled areas representing activity intensity; and (D) Time-series plots display daily cough event counts for each sensor group, with the *x*-axis representing experimental days relative to infection (Day 0) and the *y*-axis showing event frequency. Lines indicate daily event trajectories. Red dashed lines denote group-specific alarm thresholds, calculated as $\mu + 2\sigma$. Red points highlight alarm-positive days where event counts exceeded the threshold, and "Alarm" labels annotate these events. All panels share a fixed *y*-axis scale to facilitate cross-group comparison. Groups are those defined in Table 1.

cators of respiratory compromise prior to the onset of clinically scorable symptoms (Fig. 3D).

As previously noted, the alarm threshold was exceeded during the initial days of the experiment, particularly on Day –9, across multiple groups. These early activations occurred prior to infection and baseline calibration, and all animals tested negative for major respiratory pathogens upon arrival. This suggests that the elevated cough activity was likely non-infectious and instead associated with management-related factors.

Discussion

Early detection of respiratory disorders in calves is critical for managing BRDC, yet conventional clinical scoring methods often detect illness only after substantial physiological compromise has already occurred. In this study, we demonstrate that microphone-based, AI-powered acoustic monitoring provides a sensitive, automated, real-time, and non-invasive window into respiratory health, capturing cough dynamics with a temporal resolution far exceeding that achievable through manual observation. By continuously monitoring calves experimentally co-infected with Influenza D virus, Bovine Coronavirus, and Mycoplasma bovis, the system revealed a clear, biologically coherent trajectory of cough expression that emerged around Day +5, peaked near Day +11, and declined thereafter. This pattern aligns with the established pathogenesis of BRDC, in which early viral infection primes the respiratory tract for subsequent bacterial colonisation and clinical exacerbation (Cummings et al., 2022; Gaudino et al., 2022a; Griffin et al., 2010; Salem et al., 2019).

Importantly, the system also captured the therapeutic impact of preinfection tilmicosin, with treated calves showing a marked reduction in cough frequency from Day 9 onward. This dual capability positions microphone-based monitoring not only as an early-warning tool but also as a valuable instrument for evaluating treatment efficacy and supporting evidence-based antimicrobial stewardship.

Spatial analysis further validated the acoustic system's robustness. Each microphone reliably captured pen-specific coughs with minimal cross-detection and classification confidence exceeding 90%, demonstrating strong localisation capacity despite the inherently noisy barn environment. This precision is essential for commercial settings, where multiple pens, high stocking densities, and overlapping sound sources can complicate signal attribution. Notably, the system anticipated the rise in clinical symptomatology by approximately two days, an early-warning advantage consistent with findings from other precision livestock farming technologies such as accelerometers and smart feeders, which detect behavioural deviations prior to overt clinical signs (Bushby et al., 2024; Kim et al., 2024). The integration of a cough-frequency-based alarm system operationalised this early-warning potential, with alarms triggered up to 2 days before peak clinical scores, particularly in the infected and untreated group. Although early acoustic deviations can help identify pens that may require closer attention, an alarm should not be interpreted as a diagnosis or a direct indication for treatment. In practice, such alerts would simply prompt a closer clinical examination rather than automatic intervention, enabling individualised treatment of affected animals to reduce disease symptoms in carriers and prevent transmission to healthier animals. Whether this ultimately reduces antimicrobial use, workload, or disease prevalence and severity depends on how farmers and veterinarians integrate the system into their routine. At present, there are not enough scientific studies to support the evidence that an acoustic alarm system alone can improve (or has the potential to) improve health or welfare; their primary value lies in supporting earlier recognition and timely treatment, while serving as a support rather than a replacement for targeted human evaluation of animals in the initial stages of illness. Demonstrating concrete benefits will require validation under commercial farm conditions, where background noise, management activities, and environmental variability may affect system performance.

From a technical perspective, our approach builds on and extends previous work in bovine acoustic monitoring (Carpentier et al., 2018; Kim et al., 2024; Vandermeulen et al., 2016). Earlier studies demonstrated promising performance for calf-cough detection using handcrafted acoustic features or complex model architectures, typically achieving precision and recall in the 0.80–0.90 range. Our HuBERT-based pipeline achieved comparable or superior performance while providing a more streamlined, scalable framework. HuBERT, originally developed for learning representations of human speech, captures rich hierarchical acoustic features (Hsu et al., 2021). Its ability to encode subtle temporal and spectral patterns proved advantageous for distinguishing coughs from diverse barn noises, even under challenging acoustic conditions. The use of short, non-overlapping 1-second windows enabled fine-grained temporal tracking, while a conservative decision threshold (≥ 0.85) and notch-based noise augmentation enhanced robustness during deployment.

Despite the promising results, several limitations should be acknowledged. The study was conducted with a limited number of calves and a restricted set of pathogens, which may constrain the generalisability of the findings across different herds, production systems, and epidemiological contexts. The housing configuration and microphone placement were optimised for this experimental setup, and acoustic propagation may differ in larger or more heterogeneous farm environments. In particular, commercial farm conditions involve substantial background noise from routine management activities such as feeding, regrouping, cleaning, and equipment use, as well as additional sound sources associated with solid-feed consumption. These factors may influence the performance of acoustic monitoring as an on-field alert system and should be evaluated in commercial settings.

Another limitation is that the system operates at the pen level and cannot attribute coughs to individual animals. While this is appropriate for early detection of group-level anomalies, it limits its diagnostic resolution in mixed-health groups. Although the HuBERT architecture has shown strong performance on human acoustic tasks, its application to livestock remains novel, and further benchmarking against alternative self-supervised models is needed. Future studies should integrate environmental and behavioural data, including air quality metrics, activity patterns, and feeding behaviour, to better disentangle pathological signals from management-related noise.

Finally, cough sounds contain latent acoustic features, such as duration, peak frequency, and spectral shape that may encode clinically relevant information, including cough types and associated vocalisations (Ijaz et al., 2022). These traits could support the classification of cough types and provide deeper insight into health and welfare status, potentially distinguishing physiological coughs associated with disease severity, discomfort, or distress. To fully harness this diagnostic potential, future AI-acoustic systems should move beyond binary detection and incorporate richer acoustic profiling capable of supporting more informed decisions regarding antimicrobial use and management interventions.

Conclusion

Overall, our findings demonstrate that microphone-based acoustic monitoring is a powerful and practical tool for early detection of respiratory disease in calves, offering continuous, objective, and high-resolution surveillance that complements and enhances

traditional clinical assessment. The strong performance of the HuBERT-based classifier, combined with the system's ability to anticipate clinical symptomology and quantify treatment response, underscores the potential of acoustic precision livestock farming technologies to enhance respiratory health management. As farms continue to scale and labour availability declines, such automated monitoring systems will become increasingly indispensable for safeguarding animal health and welfare, reducing antimicrobial use, and improving the sustainability of livestock systems.

Supplementary material

Supplementary Material for this article (https://doi.org/10.1016/j.animal.2026.101928) can be found at the foot of the online page, in the Appendix section.

Ethics approval

The local animal care and use committee and the French Minister of Research reviewed and approved the study protocol for the calves' study (reference: APAFIS #49942-2024062000479999 v3, dated 23 September 2024, Ethics Committee no. 115). All protocols were conducted in accordance with the EEC Regulation (No. 2010/63/UE) governing the care and use of laboratory animals, which has been in effect in France since 01 January 2013.

Data and model availability statement

Examples of raw acoustic recordings, preprocessed datasets, and the model checkpoint used in this study are available via the INRAE data repository (https://entrepot.recherche.data.gouv.fr/dataverse/inrae), under the DOI [https://doi.org/10.57745/FP5LA9].

The repository includes four raw audio files, two from each of two nights, recorded at the ENVT experimental facility using directional microphones. Each file spans approximately 6–7 h and reflects the unprocessed acoustic environment of the pens during nighttime recording sessions. In addition, the dataset also includes a curated subset of 40 short audio clips: 20 manually annotated cough events and 20 non-cough events, representing a range of cough events and barn sounds. These clips were used to train and evaluate the detection algorithms. The repository also contains a HuBERT-based classifier checkpoint (.pth) saved during early training iterations (no training or inference implementation is included). All materials are provided for research and educational purposes only. Users are encouraged to cite the corresponding publication and acknowledge the authors when using these resources.

For access to the full event-level dataset or further technical details, interested researchers may contact Ignasi Nou-Plana at: ignasi.nou@salle.url.edu.

Declaration of generative AI and AI-assisted technologies in the writing process

Generative AI tools, including Microsoft Copilot, were used to assist with editing and refining the manuscript. The authors reviewed and approved all content to ensure accuracy and scientific integrity.

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Declaration of interest

The authors declare no competing financial or personal interests that could have influenced the work reported in this paper.

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